Inversion of a New Designed ANN-Based 3-D-RTM Emulator by Continuous MCMC Technique to Monitor Crop Biophysical Properties Using Sentinel-2 Images

Achraf Makhloufi and Abdelaziz Kallel

***Abstract*— Precise monitoring of the crop growth is useful for agriculture management systems that allow to improve and sustain food security. Measuring the biophysical properties, such as leaf area index (LAI) and chlorophyll (Cab) content, is the key to properly following the growth cycle of crops. In this perspective, we propose an innovative approach for monitoring landscape-scale biophysical properties of wheat and barley field crops using Sentinel-2 time-series imagery based on accurate 3-D direct and inverse radiative transfer modeling. It relies on an original 3-D radiative transfer models (RTM) emulator architecture based on a residual artificial neural network (ResNet). We employed the well-known discrete anisotropic radiative transfer (DART) model to simulate Sentinel-2 images and generate a database that serves to train and validate the ResNet emulator. To do it, two realistic mockups, both composed of soil and 3-D wheat plants, are set up. The first mockup mimics the wheat and barley from their start of growth to their full growth, and the second reproduces the features of the wheat and barley plant when they turn yellowish and mature for harvesting. Besides, we extract the bare soil spectra signature from Sentinel-2 images acquired before sowing. Results of the proposed emulator show its similarity to DART simulation while it is much faster. These emulator features make it possible to adapt an inversion technique based on the Markov Chain Monte Carlo (MCMC), specifically, as a novelty, a continuous MCMC. The latter exploits the emulator to perform the iterative sampling processing on all continuous values in allowed intervals according to each sought biophysical property until converging to its respective posterior distribution. The proposed scheme of RTM emulator inversion based on continuous MCMC and using Sentinel-2 imagery is called "REMI." It ensures high retrieval performances, which reach an RMSE equal to 0.06 and 0.1 for LAI and Cab, respectively.**

***Index Terms*— Artificial neuronal network (ANN), crop biophysical properties monitoring, emulator, MCMC, sentinel-2 time series.**

I. Introduction

WHEAT and barley are strategic raw materials that have played a central role in the development of societies and are among the key elements of food security. The demand for these products is globalizing and increasing constantly. It is, therefore, crucial to monitor these cereals' growth and health to ensure an optimal yield crop in terms of quantity and quality. This can be achieved by monitoring the cereals' biophysical properties, such as chlorophyll (Cab) and leaf area index (LAI), throughout the growing period [1]. In this context, remote sensing has proven its performance mainly using satellite images with high spatial and temporal resolution [2], [3], [4]. Various studies [2], [3], [5] have shown the effectiveness of Sentinel-2 A and B satellites in learning about vegetation biophysical properties; moreover, to retrieve these biophysical properties from the remotely sensed reflectance observation, a relationship between them and their relative spectral signatures must be established. It can be done through semi-empirical techniques that rely on relationships between vegetation indices and sensed satellite data or the radiative transfer models (RTM). In the first case, the semi-empirical methods remain dependent on the studied field because it requires empirical parameter calibration using a number of field data [6]. In the second case, physical models, RTM, can overcome this issue by reproducing the physical phenomena, such as sunlight's absorption and scattering by surfaces and volumes [7], [8], [9].

RTMs are distinguished primarily by their modes of modeling the observed scene in 1-D or 3-D [10]. In the first case, the scene is simplified to superimposing homogeneous layers with different physical properties; for example, the canopy is modeled by a turbid medium, which is horizontally homogeneous and vertically can be as variable as the different canopy layers [8]. Scattering by arbitrary inclined leaves (SAIL) is a well-known and widely used 1-D canopy reflectance model [8]. Later, the leaf scattering optical model PROSPECT [17], which allows linking the leaf reflectance and transmittance to its biophysical properties (Cab content, water content, mesophyll structure, etc.), has been integrated to SAIL to produce PROSAIL [10]. Although they have shown its simplicity and efficiency for homogeneous scenes, it neglects some aspects of the real 3-D architecture of the landscape. For instance, they are quickly saturated when trying to simulate high LAI values [11]. For instance, when LAI increases, the reflectance increases in the near-infrared (NIR) due to the multiple scattering by leaves, which are highly diffusing. When the medium is homogeneous, the signal saturates after

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a certain value of LAI since the leaves in the lowest layer are completely hidden by the leaf at the top. This is not usually the case since the vegetation is generally clumped; for instance, tree leaves belong to the crown; in this way, sun radiation can directly reach the leaves at the bottom and the reflectance continues to increase in NIR for high values of LAI [11]. In this sense, 3-D RTMs are more sophisticated in their scene representation since they can reproduce a 3-D mockup architecture that mimics the real observed scene [9]. For instance, the discrete anisotropic radiative transfer (DART) [9], [12], [13] offers a great variety of element choices (e.g., turbid or geometric facets, woody elements, 3-D trees mockups, etc.) to mimic the real scene. Besides, it simulates the radiation's propagation in all the overlapping elements that made up the complex 3-D mockup and all associated physical mechanisms by tracking the radiation propagation and scattering using the so-called "Monte Carlo ray tracing" [18]. This precision is, however, at the cost of the slowness of the process due to the time required for ray tracing. To bypass this bottleneck, new methods based on machine learning with low computation time are developed to approximate the RTMs called "emulators" [14]. RTM's data simulations are exploited to learn the emulator by using artificial intelligence techniques such as neural networks [15]. After the learning stage, the emulator will be able to well approximate the RTM simulation in a much faster time while neglecting the effect of the variables' number.

To inverse RTM, traditionally, the latter produces the so-called "lookup table" (LUT), composed of samples of all the possible combinations of the biophysical properties and variables used in the setting model with their simulated signatures. An exhaustive search in the LUT permit retrieval of the biophysical properties of the crop by finding the closest signature to each given remotely sensed pixel [34]. LUT is a simple method for inversion, but it depends directly on the dataset sample numbers that are generated by the RTM, which makes it sometimes limited in terms of accuracy, mainly when the number of unknown variables is too much. In the case of a fast simulator, it is possible to use more sophisticated inversion techniques. For instance, several approaches exploit artificial neuronal networks (ANN) in the inversion stage. Bacour et al. [38] use RTM 1-D PROSAIL to produce an LUT that serves to feed an ANN to learn the relationship between reflectances and the required biophysical parameters. After the learning stage, the proposed ANN will behave as an inverse model, and it will estimate crop's biophysical properties from a given reflectance. To overcome 3-D RTM slowness, Makhloufi et al. [2] precompute a simulation database of DART to feed its designed ANN, and consequently, they expect more accuracy in the inversion stage than LUT. Chaabouni et al. [16] suggest a variational method to solve the PROSAIL inversion problem by minimizing a cost function in the continuous domain of variation. In addition to the need of a fast simulator, the major inconvenience is the risk of falling on a local minimum [16]. As an alternative, Bayesian models, in particular, the Markov Chain Monte Carlo (MCMC) [3], [39], [40], lead to a global minimum in resolving the problem, thanks to the formulation of a priori knowledge and the introduction of a posteriori probability of the unknown variables in their computational model. Until now, only its application in the discrete domain case (using a precalculated database of simulations) was tested in 3-D RTM inversion since the need for running the latter several times makes the inversion impractical [3]. To overcome this issue, we design in our work, to the best of our knowledge, the first emulator of a 3-D RTM. It is based on an artificial neural network making it too fast. In turn, the latter is the key to our approach's second originality. Our suggested 3-D emulator acting as a direct model will enable us to speed up the simulation, thereby, allowing us to employ the continuous MCMC technique for inversion. Hence, the continuous MCMC will explore all the possibilities of inversion in the variation intervals of all the sought biophysical properties.

In the present study, the objective is to monitor LAI and Cab properties of wheat and barley planted areas at a resolution of 10 m using Sentinel-2 sensors. To retrieve the biophysical properties, we propose to couple a DART RTM emulator to an MCMC Bayesian model and use Sentinel-2 imagery. The model is called "RTM emulator inversion based on continues MCMC and using Sentinel-2 imagery (REMI)."

II. Materials

A. Study Sites

The study areas are spread over four sites in Tunisia, as shown in Fig. 1. There are two types of cereal: wheat and barley. As illustrated in Fig. 1(b) and (d), there are two wheat fields located in Tunis with an area of 6.2 and 2.1 ha, respectively. The third parcel is also a wheat field in Beja, with an area of 22.3 ha. The last parcel is in Zaghouan, and it is made up of barley crops with an area of 2.9 ha.

B. Satellite Data

The remotely sensed reflectance observations are ensured by Sentinel-2 time series over the crop fields. Two satellites, Sentinel-2-A and Sentinel-2-B, are in orbit offering together observations every 5 days. The sensors have resolutions of 10, 20, and 60 m with 13 spectral bands distributed in visible, NIR, and shortwave-infrared optical ranges. The 10 and 20 m bands (ten bands) are well adapted to study the vegetation spectra [19]. The Sentinel-2 images are downloaded from THEIA[1] at level-2.[2] This product is corrected from atmospheric effects by the MAJA2 algorithm [31].

C. In Situ Measurements

To collect useful measurements, it is important to understand the growth cycle of the studied crops. Barley and wheat have a similar growth shape, which is mainly divided into three stages. The first stage called "tilling starts," in general, is in January, when the seed starts to grow, and the first shoots appear from the ground. Until March, the shoots continue to

[1]The THEIA Land Data Centre is a French inter-agency designed to promote the use of satellite data: https://labo.obs-mip.fr/multitemp/a-new-french-thematic-platform-dedicated-to-land-surfaces/

[2]Level-2: Level 2 products provide surface reflectances (images) corrected for atmospheric effects.

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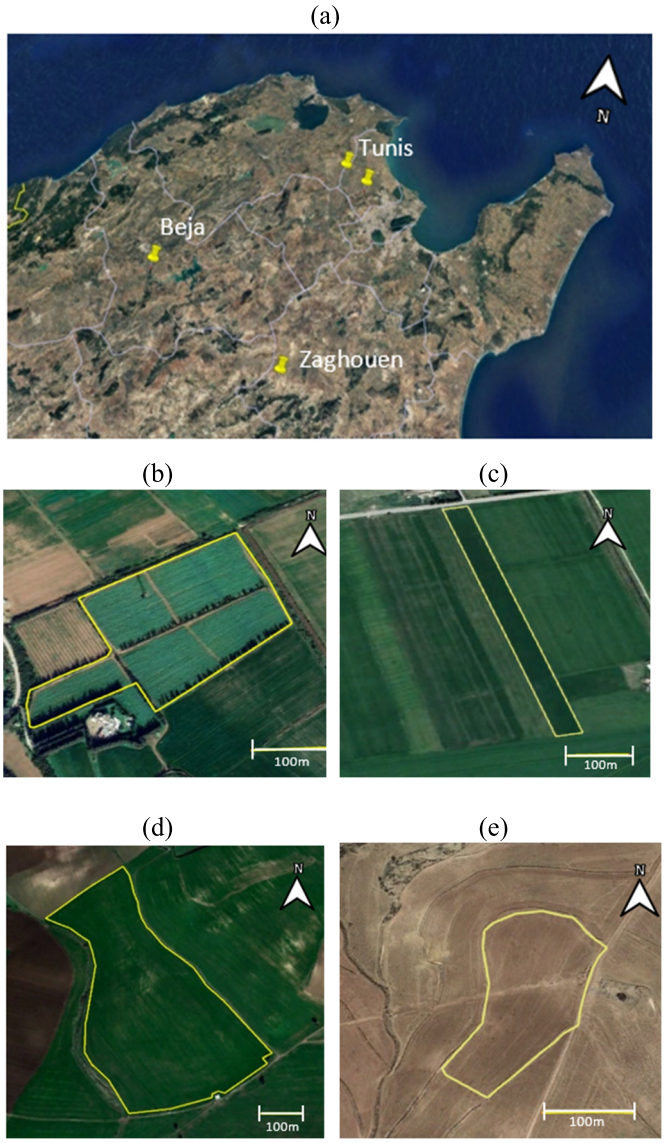

Fig. 1. Sites: (a) top right corner: 37°10′56.39″N, 10°32′45.85″E, localization in north Tunisia map; (b) top right corner: 36°55′29.01″N, 10°5′26.88″E; (c) top right corner: 36°59′8.52″N, 10°0′10.44″E, are two wheat fields in Tunis of areas 6.2 (Tun1) and 2.1 ha (Tun2), respectively; (d) top right corner: 36°40′13.12″N, 9°13′1.18″E, wheat field in Beja (Bej), 22.3 ha; and (e) top right corner 36′19′17″76″N, 9′45′21″97″ E, barley field in Zaghouen (Zag), 2.9 ha.

grow, forming leaves and the head. The latter grows from the top of the stem, and the last leaf is wrapped around it. The end of March and the beginning of April mark the second stage, where the head of the crops is full-grown, and the last leaf unwraps from around. At the end of its growth cycle, the crop ripens from green to a golden color, and it is ready to be harvested by the end of June. Based on this cycle, study sites were visited on April 12, 2021 when the crop is full-grown, and on May 21, 2021 when the crop ripens and changes color. Fig. 2 plots an example of a field measurement location. The in situ measurements involved mainly: LAI of the crops and leaf Cab.

The LAI is determined by computing half of the total area of the leaves of plants over the area of soil that these plants cover [41]. This can be possible using camera photos and based on

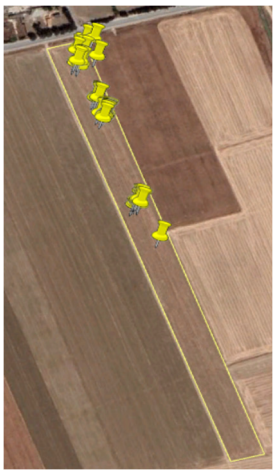

Fig. 2. Field measurements' location: Tunis crop parcel (Tun2).

the method presented in [3]. The camera, in this case, is placed below the vegetation and observing the sky. In this case, the more the sky is observed the lesser the vegetation density. The sky observation is the vegetation transmittance T. The relationship between T and LAI is given by the Beer–Lambert law [19]

$$T = \frac{I}{I_T} = \exp(-k(\text{LAI})) \tag{1}$$

where I is the sky area observed by the sensor, I_T is the total observed area, and k is the extinction coefficient, depending on the leaf angle distribution [8]. This ratio is considered as the gap fraction at ground level [21]. This later can be calculated from images taken by a camera (e.g., smartphone) from the bottom of the plant, as shown in Fig. 3(a) and (b). Then, to separate the foliage from the areas of gaps (i.e., sky) in the image [Fig. 3(c) and (d)], a classifier algorithm called "Gaussian mixture models" (GMMs) is applied. GMM is a weighted sum of Gaussian component densities that represents a parametric probability density function [35]. It has been shown to be a simple and useful tool for classifying and segmenting images [36]. The gap fraction is then computed by dividing the number of gap pixels (N_{gap}) by the total number of pixels (N_T), which are proportional to the gap fraction (I) and (IT), respectively. Thus, LAI is finally estimated using the following equation:

$$\text{LAI} = \frac{-1}{k}\ln\left(\frac{N_{\text{gap}}}{N_T}\right). \tag{2}$$

Our model assumes that leaves are characterized by randomly distributed foliage and a spherical leaf inclination angle, where $k = 0.5$ [22].

The Cab measurement on cereal fields is taken by a Cab meter handheld device: Konica Minolta SPAD-502 (SPAD) [23]. The SPAD readings were converted into leaf Cab contents by applying the coefficients of Lichtenthaler [24].

III. Methodology

In our approach, we aim to monitor the growth cycle of wheat and barley crops, specifically through the LAI and Cab

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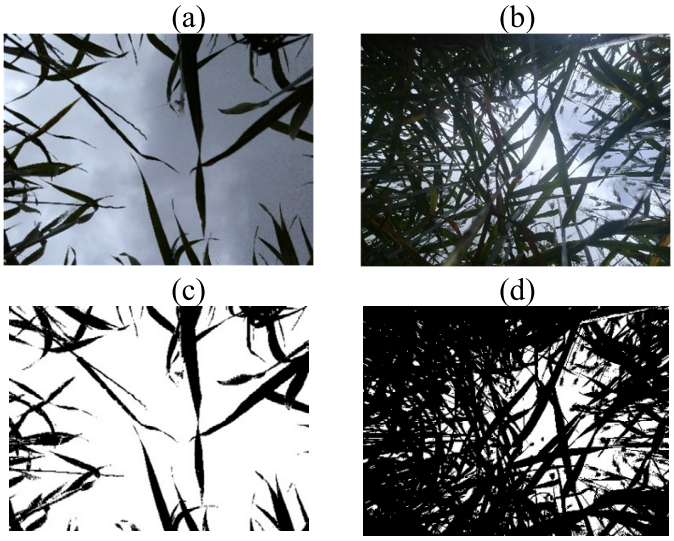

Fig. 3. Classification of images from the bottom of the plant to the sky taken by a smartphone. (a) Low LAI. (b) High LAI. (c) GMM classification applied to the image (a). (d) GMM classification applied to the image (b).

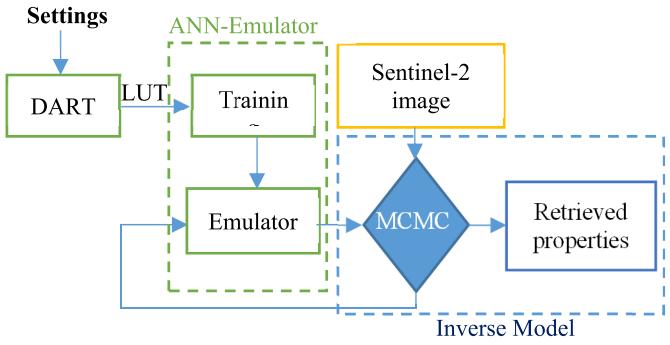

Fig. 4. Design diagram: main steps of our property retrieval approach.

variations. These biophysical parameters are obtained by the inversion of DART using the Sentinel-2 image time series. As DART is time-consuming, it will be substituted by an emulator in the inversion task to accelerate the process. The design of our approach is summarized in Fig. 4. The DART RTM is parametrized to generate an LUT composed by a reflectance simulation database corresponding to plausible combinations of the different biophysical properties (e.g., LAI, Cab, water content, brown pigments, etc.). Next, the generated LUT is used to learn an ANN. After the learning stage, this ANN will be able to act as a fast emulator of DART. Subsequently, this emulator will receive input data from the MCMC process, as illustrated in Fig. 4. More precisely, whenever a random sampling is carried out for a given biophysical parameter (e.g., Cab or LAI), MCMC employs the emulator to generate the corresponding reflectances. The MCMC algorithm then decides whether to accept or reject the proposed parameter based on the emulator's output.

A. DART Radiative Transfer Model Settings

DART is an accurate 3-D RTM that permits the simulation of Sentinel-2 observations; however, it requires rigorous parameterization to achieve realistic results. Indeed, the

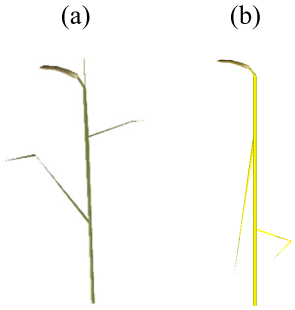

Fig. 5. Crop mockups. (a) Representative model of the first phase of growth. (b) Representative model of the last phase of growth.

mockup settings should clone the real scene, for instance, it should consider the most important effects like leaf biophysical properties, soil brightness, vegetation structure, as well as the atmospheric conditions (e.g., aerosol density). Besides, various sun orientation must be considered to have a representative data in the different year months.

Wheat and barley have almost similar phenologies and morphologies [42], [43]. To model them, we use the same 3-D mockups as the plants are similar. The differences are in the growth stage, the size, and the density of plants per square meter. For that, these parameters are put free. In addition to the structure, the biophysical properties of the leaf variation ranges are large, allowing it to cover all the possible combinations. In the following, we present the various settings that are explored in our approach.

1) Mockup 3-D Architecture: As mentioned in Section II-C, the vegetation under study goes through three phases: the tilling phase, the full-grown phase, and the harvesting phase, where the crop turns to golden color. In the two first stages, the crops have almost the same shape and color, principally growing green leaves. But in the last stage, the crop's leaves curl and turn to golden color. At this time, the brown pigments start to appear, testifying to the Cab deficiency. For these reasons, we considered two mockup models, one for both the first and second stages and another for the last stage of growth.

The mockup is 4 × 4 m planted with a representative wheat/barley 3-D plant, as illustrated in Fig. 5. In the first mockup, we used a DART database mockup of the 3-D wheat plant, whereas, in the second, the shape of the 3-D plant is adjusted specially to obtain the curled leaves shapes. This task is achieved using the software Blender (https://www.blender.org/). In our work, the assumption of a particular row structure for crops is eliminated, as during its sowing, the wheat and barley are scattered in a randomized pattern. In particular, the field measurements conducted in March and April revealed a variation in crop density ranging from approximately 300 to 500 plants/m^2, which induces several levels of clumping. Although several studies have demonstrated that varying crop row structures or levels of clumping can result in diverse spectral signatures, in our work, this variation translates into a variation in LAI. Indeed, based on our experiments, we found out that the combination of high plant density levels (i.e., $>$200 plants/m^2) and a randomized sowing pattern resulted in similar spectral signatures when

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TABLE I
CROP PROPERTY SETTINGS IN DART

	Parameters	Value	Units
θ_1	Leaf chlorophyll content (Cab)	1 7 15 20 40 60	µg cm^{-2}
θ_2	Leaf area index (LAI)	0.41 1.65 3.2	m^2 m^{-2}
θ_3	Leaf equivalent water thickness (EWT)	0.005 0.01 0.03 0.05	g cm^{-2}
θ_4	Leaf mesophyll structure (N)	1.2 1.5 1.8	unitless
θ_5	Carotenoid (Car)	10 20 30	µg cm^{-2}
θ_6	Brown pigment (Br)	0 0.5 1 2	unitless
θ_7	Leaf mass per area (Lm)	0 0.01	µg cm^{-2}
θ_8	Soil brightness coefficient (Sbr)	0.8 1 1.2	unitless

either increasing the size of the plants or the number of plants, aiming to achieve the same LAI. For this reason, the mockups were configured with the same density of planting while the growth stage remained variable. Therefore, the variation in planting density in the mockups is implicitly deduced from the value of LAI. Furthermore, knowing that stems at a given stage of growth may have different lengths. To account for this variability, lengths observed in the field are assumed to follow a normal distribution. Thus, in our mockup, stem length is randomly chosen for each plant according to a normal distribution with a given mean and a standard deviation of 10% of the mean, reflecting realistic conditions. Besides, to mimic real plantation, each plant mockup undergoes random rotation on its vertical axis. This makes it possible to have a more natural distribution of leaves and lets it possible to simulate the actual effect of sunlight transmittance, absorption, and scattering by thin superimposed leaves. We note that such effects induce high leaf interception and scattering, and as a consequence, they increase the reflectance in the red edge and NIR domain (e.g., bands B5, B6, B7, B8, and B8a of Sentinel-2).

2) Biophysical and Atmosphere Variables:

a) Biophysical properties: Leaf optical properties were simulated with the prospect model [26] on the respective ranges according to previous studies on the same crop [26], [27]. Table I summarize the considered parameters $\theta_i (i = 1, 2, \ldots, 8)$ and their variation ranges.

b) Soil brightness: The impact of soil brightness on the canopy spectrum during a satellite observation is proportional to the degree of ground visibility. In the case of the studied crop, in the full growth stage, the soil is almost covered and without significant influence on the spectrum; however, in the tilling phase and until harvest time, the crop leaves curl, and they consequently make the soil more visible, and therefore, more influential on the resulting spectrum. To consider the soil effects, we extract the soil reflectance of each studied field from Sentinel-2 images separately and as follows. First, the considered Sentinel-2 images in this processing phase are those of August or September. In fact, during this period, the soil is often bare since the harvest is already completed by July, and the next sowing season will not begin until October.

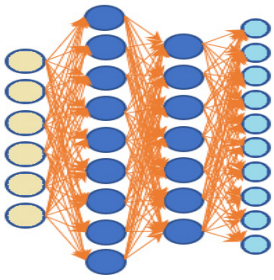

Fig. 6. Plain ANN.

Second, normalized difference vegetation index (NDVI) is computed for each pixel, and then thresholding is applied to extract bare soil and avoid pixels with a mixture of soil and possibly grass. Finally, the obtained soil spectrum is scaled to cover all possible soil reflectance throughout the year. Indeed, the soil becomes brighter, and so its reflectance increases when it is dry, for example, during the summer period, whereas it decreases when it is moist; for example, after a rainy episode it becomes darker; moreover, the soil reflectance decreases with the sun's zenith angle due to the shadowing effect. Assuming that changes in soil humidity affect all measured bands equally, this induces a linear change in the measured soil reflectances [2]. This last observation makes it possible to create several soils by multiplying the extracted bare soil by a scale factor to get brighter or darker soils.

c) Solar angles: The spectra observations depend on both the sun and the observation angles (zenith θ, azimuth φ). As Sentinel-2 observes almost on nadir, the corresponding angles are assumed $\theta = 0$ and $\varphi = 0$, whereas the sun angles in the studied regions vary from ($\theta = 18.45$, $\varphi = 124.99$) to ($\theta = 60.17$, $\varphi = 162.81$) throughout the year. To consider this variation, one sun orientation is assumed each month. This way 12 possibilities are simulated.

d) Atmosphere setting: DART offers an extensive atmospheric database. The gas model US Standard Atmosphere, 1976 (USSTD76) and the aerosol model RURALV23 [18] both correspond well to the studied area [28].

B. ANN-Based Emulator

As shown in Fig. 4, the ANN emulator is composed of two parts, the ANN training and the emulator. The latter is an already learned version of the ANN. In the first part, the ANN exploits the produced DART database. In total, 70% of the simulations are used to learn the ANN and find the relationships between the biophysical properties and solar angles as input and their corresponding simulated reflectances as output. The remaining 30% is used in the ANN validation stage. In the second part, the emulator will play an RTM role while ensuring an almost equivalent precision, in return, with low computation time.

Fig. 6 illustrates a plain ANN architecture, which is defined by a first layer depending on input parameters followed by hidden layers leading to the outputs. This kind of architectures could achieve acceptable performance for relatively simple problem but remains limited. In fact, although when increasing

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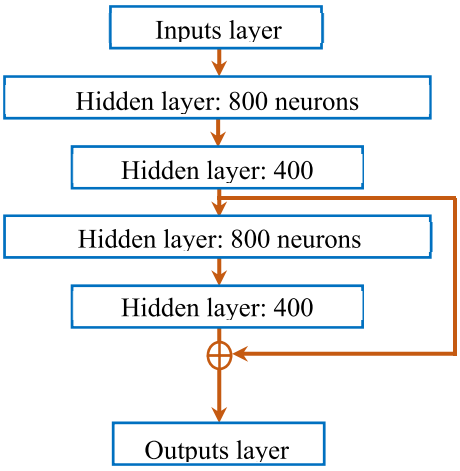

Fig. 7. Proposed architecture for the developed ANN emulator.

the number of hidden layers in the hope of further performance enhancement, the network behaves sometimes inversely, and it leads to a higher training error [29]. To overcome this issue, the so-called "residual neuronal networks" have been developed, they allow training in much deeper networks and to catch the nonlinear variable variations [29]. Their architecture consists in skipping connections, or shortcuts to jump over some layers to the outputs of an upstream layer. This leads to learning residual functions with reference to the downstream layer. This makes it possible to train deeper networks than it was possible with plain ANN, allowing consequently higher performances for complex and nonlinear problems.

Fig. 7 illustrates our proposed residual network (ResNet). It is composed of four hidden layers, where the outputs of the second layer are added back to the last hidden layer. A ReLU activation function was used on all the ANN layers except the last one where a sigmoid function was deployed to have outputs with the finest variation. An Adam optimizer with a learning rate of 10^{-4} was employed with a mean squared error (MSE) loss function [30]. TensorFlow and Keras were used to build the model. The inputs/outputs of the ANN consisting of 69 984 simulations are inherited from DART database as follows:

1) Inputs:
   i) biophysical properties: Cab, LAI, Cw, N, Br, Lm;
   ii) solar angle; and
   iii) soils reflectances at the Sentinel-2 bands (Sbr).
2) Outputs: corresponding simulated Sentinel-2 reflectances.

C. Inverse Model

LUT is the simplest way to inverse a RTM over the real observations. It consists of taking finite samples state of each unknown biophysical property (θ_i) (Table I) and generating relative observations, then finding within the simulated data the nearest signature to the real observation. This way, LUT remains dependent on the finite number of sampling steps relating to each simulated θ_i. Otherwise, sampling directly from our continuous target distribution $P(\theta_i)$ should fit the actual observation with higher accuracy. Nevertheless, this

TABLE II
$\{\theta_i\}_{i=1,\ldots,8}$ SPACE OF VARIATION

	Parameters	Value	Units
θ_1	Leaf chlorophyll content (Cab)	[1 : 90]	µg cm^{-2}
θ_2	Leaf area index (LAI)	[0.1 : 4]	m^2 m^{-2}
θ_3	Leaf equivalent water thickness (EWT)	[0.001 : 0.05]	g cm^{-2}
θ_4	Leaf mesophyll structure (N)	[1:2]	unitless
θ_5	Carotenoid (Car)	[5:40]	µg cm^{-2}
θ_6	Brown pigment (Br)	[0 : 2]	unitless
θ_7	Leaf mass (Lm)	[0 : 0.01]	µg cm^{-2}
θ_8	Soil brightness coefficient (Sbr)	[0.8 : 1.5]	unitless

requires computing the a posteriori probability of the different parameters. Bayesian statistic approaches like the MCMC can perform such an estimation [3], [39], [40]. Indeed, MCMC is an iterative processing of sampling that can converge to the a posteriori distribution. To apply MCMC in our approach, each element $\{\theta_i\}_{i=1,2,\ldots,8}$ is assumed to follow a priori a uniform distribution (3) in their respective space of variation defined in Table II. Such a distribution allows for all possible values to be sampled with equal probability, which is important when trying to explore the entire parameter space. So unlike LUT method, θ_i are not anymore discrete finite state sets as seen in Table I

$$\theta_i \sim U_{\theta_i} \tag{3}$$

where U_{θ_i} is the continuous uniform distribution over the interval of variation of θ_i.

The observation R (with dimension (10×1) corresponding to ten Sentinel-2 bands) is represented by

$$R = f(\Theta) + \varepsilon \tag{4}$$

where f is our emulator function, which will simulate an observation given $\Theta = \{\theta_i\}_{i=1,2,\ldots,8}$ a combination of the input parameters. ε models possible errors from simulation or due to observation's perturbation. It is assumed to follow a normal distribution [3]

$$\varepsilon \sim \mathcal{N}\left(0, \sigma^2 I\right) \tag{5}$$

where $\mathcal{N}$ is a multivariate Gaussian distribution with first parameter the mean and the second the covariance matrix, σ^2 is the variance of the noise and I is the identity matrix of the same dimension as the observation vector R (4).

The a posteriori probability of the different unknown θ_i is

$$P(\Theta|R) = \frac{P(R|\Theta) \times P(\Theta)}{P(R)}. \tag{6}$$

In this equation, it is first noted that the a posteriori probability is proportional to $P(R|\Theta) \times P(\Theta)$ because $P(R)$ is independent of Θ

$$\begin{aligned} P(\Theta|R) &\propto P(R|\Theta) \\ &\propto e^{-\frac{-\|R-f(\Theta)\|^2}{2\sigma^2}} \\ \text{where } \|R - f(\Theta)\|^2 &= \sum_{i=1,\ldots,10} (R_i - f(\Theta_i))^2. \end{aligned} \tag{7}$$

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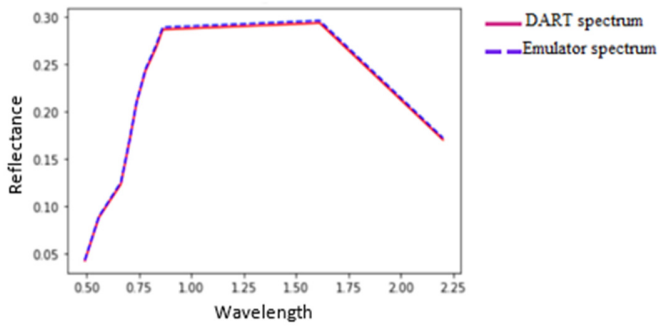

Fig. 8. DART versus emulator spectra test.

Second, since $f(\Theta)$ corresponds to the DART simulator (or its emulator), it is not linear. (7) is not a Gaussian distribution, and therefore, is not practical to directly take samples from that target probability distribution. This situation specifically motivates us to employ the Metropolis sampler [32]. Finally, given that the biophysical parameters are a priori independent, and all values are possible, we choose as proposal distribution a uniform law over the whole variation space.

Since then, a random metropolis hastings (MH) sampling is performed then the emulator produces the corresponding reflectances, finally the MH makes the choice to accept or reject the sought parameter.

IV. Results

Henceforth, the proposed scheme of RTM emulator inversion based on MCMC and using Sentinel-2 imagery is called "REMI." We propose first to study the emulator performances, then we validate the whole approach using successively simulated and actual data.

A. ANN/Emulator Performances

A neural network can be qualified as efficient when the MSE computed between the predicted data and the actual one is lower than a given threshold. Nevertheless, this can lead to overfitting the training data, and thus a difficulty in predicting new unseen data; therefore, the goal is to design an ANN that achieves optimal performance in terms of MSE while maintaining strong generalization capabilities for unknown data. Several plain ANN architectures, including our proposed Residual-ANN (ResNet), were all trained and validated using the reflectance database corresponding to the property variations presented in Table I. The latter table contains large variation intervals for all the parameters to ensure that all real combinations are taken into account. Table III summarizes their different layer structures and their corresponding training errors. The best among all tested plain ANN was composed of two hidden layers, composed of 800 and 400 neurons, respectively (Table III: A2). It achieves at the best case 3.17 $\times$ 10^{-5} MSE in learning stage. By trying to add layers to reach deeper learning, the training error starts increasing (Table III: A3–A5). Contrariwise, our proposed architecture based on ResNet, illustrated in Fig. 7, ensures more deeper learning while achieving the best training error (Table III: A6).

TABLE III

Examples of Computed MSE for Several Plain ANN Architecture (A_1–A_5) Versus Our Proposed Residual ANN (A_6). Plain ANN are Described by the Succession of Layers "L" Followed by the Neuron Number

ANN architecture	Layers/neurons	Training MSE $\times 10^{-05}$
A_1	L_1/800 L_2/100	3.4928
A_2	L_1/800 L_2/400	3.1717
A_3	L_1/800 L_2/400 L_3/100	2.6932
A_4	L_1/800 L_2/400 L_3/100/ L_4/50	3.4990
A_5	L_1/800 L_2/400 L_3/100/ L_4/50 L_5/25	3.5303
A_6	Our proposed residual ANN	0.5930

TABLE IV

DART Model Properties Settings for New Database Test

Parameters	Value	Units
Leaf chlorophyll content (Cab)	5 12	µg cm^{-2}
Leaf area index (LAI)	0.41 1.65 2.58	m^2 m^{-2}
Leaf equivalent water thickness (EWT)	0.02	g cm^{-2}
Leaf mesophyll structure (N)	1.3	unitless
Carotenoid (Car)	15	µg cm^{-2}
Brown pigment (Br)	0.75	
Leaf mass	0	

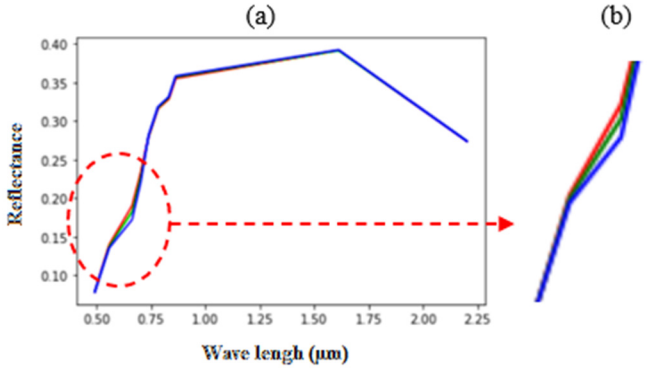

Fig. 9. Emulator spectra test. (a) Three predicted emulator spectra by varying Cab (μg cm^{-2}): red Cab = 3; green Cab = 5; blue Cab = 9. (b) Zoomed-in view to identify spectrum variations.

Fig. 8 illustrates an example of DART simulated spectrum (red spectrum) and our predicted emulator spectrum (dashed blue spectrum). The spectra seem superimposed testifying the emulator performance. Furthermore, our ResNet is put to the test to evaluate its behavior versus the overfitting dilemma. To this end, new data that never used neither in learning nor in validation stages was simulated in DART as summarized in Table IV. For all these new tested data, our emulator achieves a MSE less than 1.2×10^{-3}. Furthermore, Fig. 9 demonstrates an example of small Cab variations (3, 5 and 9 μg cm^{-2}) and the corresponding emulator response sensitivity. Although the new Cab values are very small, the responses of our ResNet emulator are easy to observe in the visible bands where the reflectance is too sensitive to the Cab content. This bears witness to the accuracy of our emulator.

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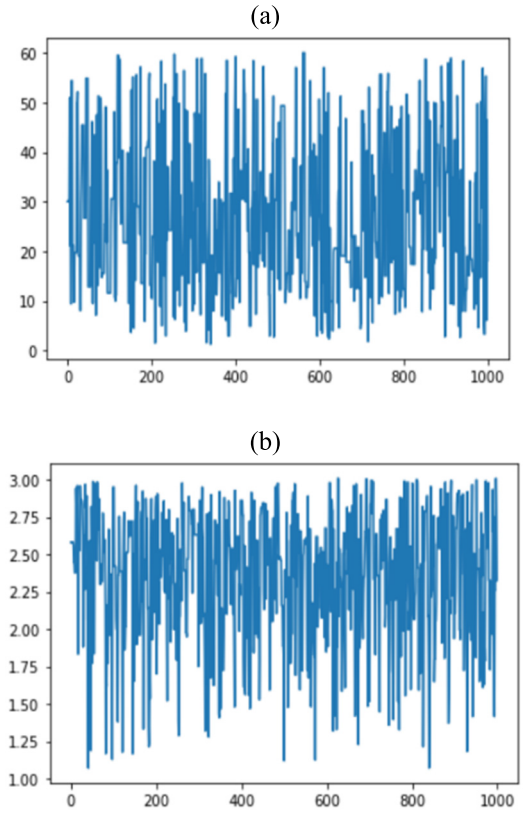

Fig. 10. A trace plot for: (a) Cab and (b) LAI.

B. Inversion Performance on Simulated Data

Before dealing with actual satellite image, it is important to validate our inversion approach, REMI, on simulated data since it allows to evaluate the performance of the emulator and the MCMC techniques without caring about the accuracy of DART nor the quality of the acquired data. As a baseline, we consider here the LUT technique with as simulated database the one used for training our DART emulator.

1) Convergence of the MCMC Algorithm: Our MCMC technique was applied using the Metropolis sampler. Checking the mixing properties and convergence of the sampled chains is crucial to guarantee a trustworthy MCMC approach. MCMC requires that the draw converges to a law of probability (a posterior distribution) after N iterations. In our case, N was fixed at 1000 iterations. It was applied to $N_s = 3338$ simulations of Cab and LAI. The values of Cab are from 5 to 60 μg cm^{-2} with step 1 μg cm^{-2} and 12 LAI values are considered in $L = \{1.06, 1.19, 1.34, 1.49, 1.65, 1.82, 2, 2.19, 2.38, 2.58, 2.79, 3.01\}$. The prediction of Cab and LAI is obtained by calculating the mean of the predictions of the last 20% of the N iterations: $N_{20} = 800$–1000. Fig. 10 reveals an example of trace plots for Cab and LAI sampling. They demonstrate that MCMC has an efficient mixing behavior by smoothly exploring the area with the highest densities respectively for Cab and LAI (resp. 5–60, and 1.06–3.01) rapidly transitioning from one point to another, which let the MCMC quickly converging to the posterior distribution. Besides, Fig. 11 illustrates the autocorrelation at various lags computed for both Cab and LAI. The plots demonstrate a rapid decay (around ten iterations) to reach zero, which suggests that the Markov chain is getting close to reaching its stationary distribution. Finally, to confirm that MCMC has converged after N iterations, we propose to show that the mean and the standard deviation remain constant even continuing the process of sampling after N. To check the convergence, one begins by computing the mean M and the standard deviation δ of N_{20} given by (8) and (9), respectively. Then, the iterations are further advanced by 20% of N: $N_{20\text{bis}} = 1000$–1200, and M and δ of this extension are similarly recalculated. Yet, RMSE between the N_{20} and $N_{20\text{bis}}$ iterations were computed using (10). Fig. 12 compared the retrieved Cab and LAI values using the N_{20} and $N_{20\text{bis}}$ iterations. The results show that the normalized RMSE (NRMSE[3]) between N_{20} and $N_{20\text{bis}}$ approaches zero. For instance, NRMSE $(M_{\text{Cab}}) = 0.015$ and NRMSE $(M_{\text{LAI}}) = 0.006$. Besides, NRMSE $(\delta_{\text{Cab}}) = 0.11$ and NRMSE $(\delta_{\text{LAI}}) = 0.08$ indicates that δ are almost stable, too. These results implies that the MCMC algorithm is well converging

$$M_j = \frac{\sum_i X_{j,i}}{200}, \quad j = 1, \ldots, N_s \tag{8}$$

$$\delta_j = \sqrt{\frac{\sum_i \left(X_{j,i} - M_j\right)^2}{200}}, \quad j = 1, \ldots, N_s \tag{9}$$

where X = Cab or LAI, i = 800–1000 for N_{20} and i = 1000–1200 for $N_{20\text{bis}}$.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{i=3338} \left(Y_{j,N20} - Y_{j,N20\text{bis}}\right)^2}{3338}} \tag{10}$$

$Y_{j,N20\text{bis}} = M_j$ (respectively δ_j) for $N_{20\text{bis}}$ iterations; $\times$ retrieved Cab (resp. LAI) of the latest 20% of N (N_{20} = 800–1000); $\cdot$ retrieved Cab (resp. LAI) of the added iterations = 20% of N ($N_{20\text{bis}} = 1000$–1200).

2) Robustness Against Noisy Observations: Generally, measurements from satellites can be affected by imperfections such as sensor noise or atmospheric effects. It is, therefore, important to see how our inversion approach behaves against these possible inaccuracies. Fig. 13 draws the performance of Cab and LAI retrieval by our approach and by LUT variations as functions of the noise (ε) standard deviation σ from 0.001 to 0.03 [cf. 4]. For a given σ, NRMSE is computed between the actual and the retrieved Cab and LAI by our approach and by LUT inversion. The values of Cab and LAI considered here are from 3 to 60 μg cm^{-2} and from 1 to 3, respectively.

It is noteworthy to observe from Fig. 13 that NRMSE and the standard deviation of MCMC for the retrieved Cab and LAI slowly increase with an almost steady level for σ range from 0.001 to 0.01 but begins to take on larger values above 0.01. This can indicate the limit from which the signal starts to lose its spectral signature due to the noise intensity that becomes highest than the useful signal. Besides, Fig. 13 shows that $\text{NRMSE}_{\text{MCMC}}$ of the retrieved Cab and LAI remain lower than $\text{NRMSE}_{\text{LUT}}$ for all the cases. Moreover, MCMC standard deviation for LAI remains lower than $\text{NRMSE}_{\text{LUT}}$, which proves the robustness of our technique against noise as it

[3]Normalized RMSE = RMSE/$(I_{\text{Max}} - I_{\text{min}})$ with I: simulated variable (Cab or LAI).

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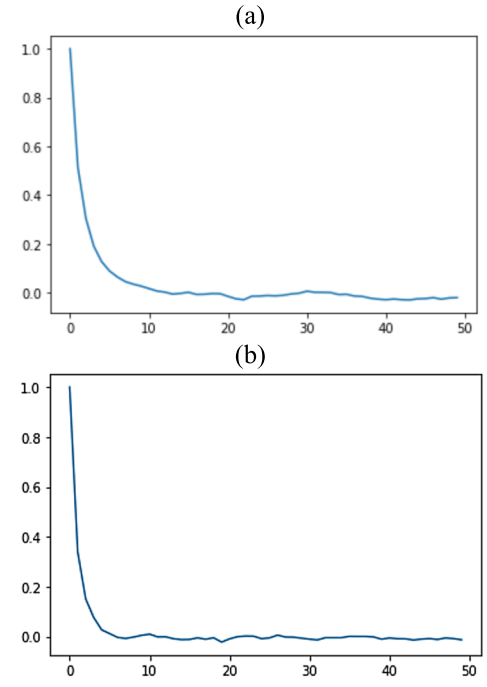

Fig. 11. Autocorrelation at various lags. (a) Cab and (b) LAI.

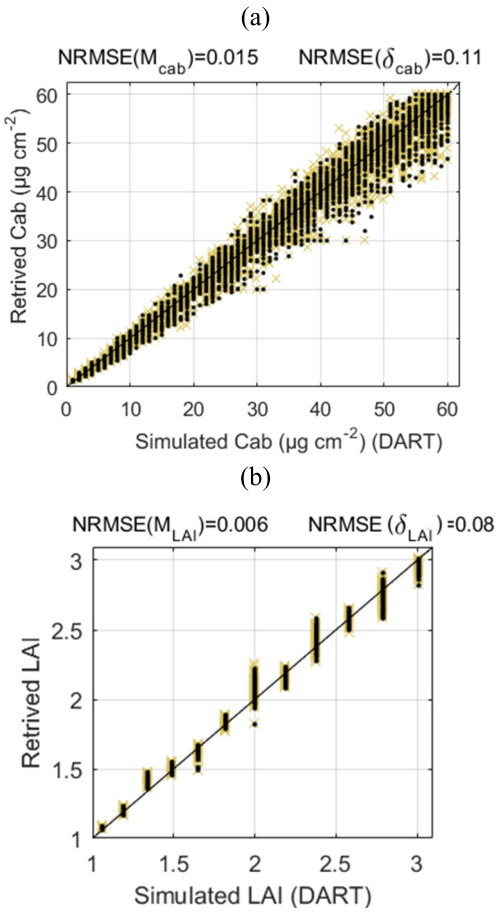

Fig. 12. MCMC convergence.

looks for the optimal solution within the whole variable space without discretization. Besides, NRMSE of Cab is generally higher than the LAI one, in addition its increase as a function of the noise is generally the fastest (except for the highest σ

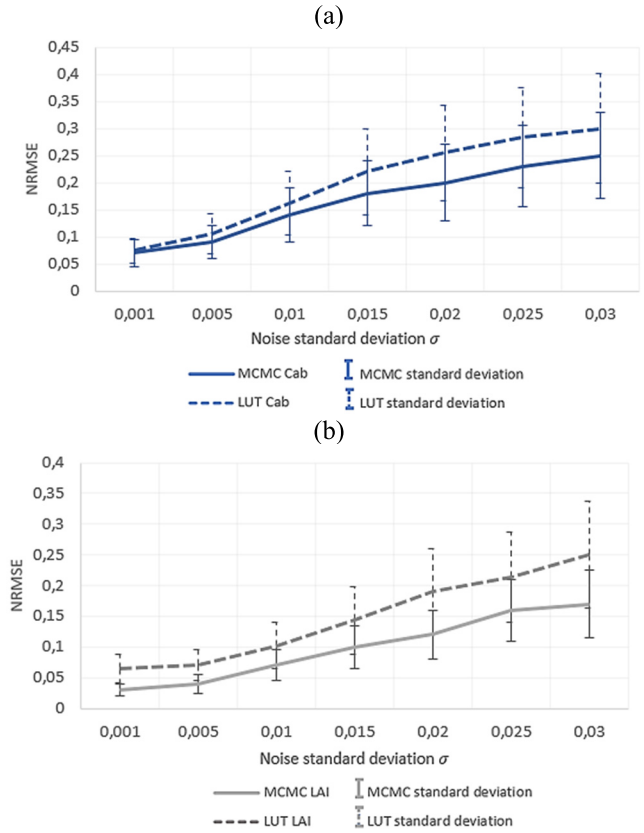

Fig. 13. Performances of our proposed technique and LUT versus noisy observation by a Gaussian signal $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. (a) NRMSE for retrieved Cab. (b) NRMSE for retrieved LAI.

where the signature is almost useless). This proves that the Cab retrieval is more difficult and more vulnerable to noise than LAI. This is explained as follows: the reflectance is sensitive to Cab variation only from a part of the observed spectrum, which is the visible and the red-edge domains; however, the sensitivity to LAI is from the entire spectrum. This explain why the Cab inversion is harder. More in depth, the increase of Cab implies a high decrease in red and blue reflectance, less decrease in green and red-edge. This implies that Cab retrieval is based on the difference between reflectances in these bands. This way a noise on the reflectance in one band or more can lead to high error in Cab content estimation. Conversely, the increase of LAI implies a global decrease of reflectance in visible and shortwave-infrared and an increase in the NIR domain, therefore the impact of noisy bands is less effective. For instance, if one band in NIR increases and another decreases then the effects will cancel each other out.

3) Inversion Performances: Given a DART simulation database with 3338 cases, we try here to retrieve the values of the actual simulated Cab and LAI using our proposed method, REMI. Fig. 14 gives an overview about the REMI performances by illustrating the histogram of the computed RMSE between the reconstruct signature using the retrieved parameters from one side and the original signature from the other. Most of the retrieved parameters ensure a small value of RMSE. Besides, the RMSE mean for all the data is less than 0.001 (cf. red line of Fig. 14). This result is important for

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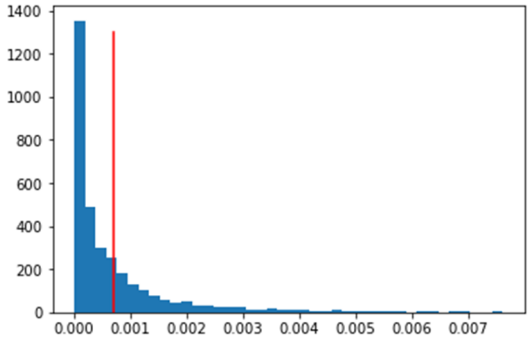

Fig. 14. REMI inversion: Histogram of RMSE between the actual and the retrieved signature. The red line corresponds to the average value.

evaluating the performance of the emulator and the MCMC technique that allow to converge to the actual signature with small error. The latter is lower than the sensor and atmospheric noise in average [33]; this means that the emulator/MCMC noise does not affect too much the inversion performances.

Fig. 15 details the results of Cab and LAI retrievals. It compares the retrieved Cab and LAI values to the actual ones. Cab varies from 3 to 60 μg cm^{-2} and LAI varies from 1 to 3. As can be identified in Fig. 15, LUT is composed of finite states, which are not optimal to represent whole the Cab and LAI variations. Contrarily, our proposed approach ensures a continuouss variation and stays close to the actual values. In addition, NRMSE reveals the accuracy of our approach where $\text{NRMSE}_{\text{MCMC}} = 0.045$ and 0.002 for Cab and LAI, respectively. Besides, they are lower than those of LUT, which are 0.1579 and 0.1215 for Cab and LAI, respectively; moreover, according to Fig. 15, the respective average REMI estimates of Cab and LAI are close to the actual one as the curves are close to the 1:1 line, which is not the case for LUT; nevertheless, Fig. 15(a) illustrates that Cab mean REMI estimation starts to little diverge from the actual ones for Cab $>$ 50 μg cm^{-2} as it is no longer too close to the 1:1 line. In fact, this phenomenon is linked to the saturation effect. The saturation of a parameter (e.g., Cab) means that its value augmentation has no more effect in the observed spectrum and consequently, this leads to significantly decrease the inversion accuracy. In our case, the simulated Cab $>$ 50 μg cm^{-2} is almost on its saturation limits. This leads to observing a tiny effect on the observed spectrum while it variates, which, in turn, decreases the inversion accuracy.

1) Retrieved Cab versus simulated Cab (a), and the mean of the retrievals (b).
2) Retrieved LAI versus simulated LAI (c), and the mean of the retrievals (d).

Fig. 16 presents the same experiments as Fig. 15 but with noise observation. It confirms the vulnerability of Cab recovery from a noisy signal compared to LAI as mentioned in Fig. 13. Although both Cab and LAI retrievals move away on either side of the 1:1 line due to the added noise, the mean error for both LAI and Cab using REMI remains close to the 1:1 line, while LUT highly increases specifically for Cab but with less impact on LAI since its inversion is more robust than Cab, as seen in Section IV-B2.

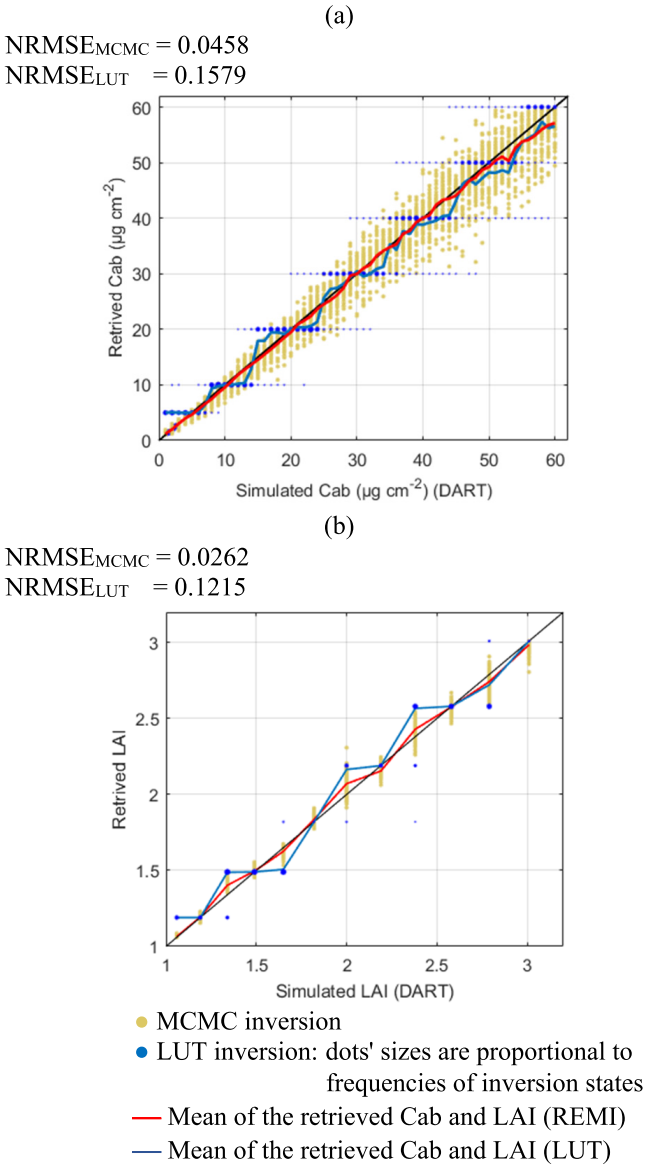

Fig. 15. REMI and LUT inversion performances in case of unnoisy observations.

C. Monitoring and Validation

1) Biophysical Proprieties Monitoring: Each Sentinel-2 image of a parcel allows to derive maps of the biophysical properties (Cab, LAI, leaf equivalent water thickness (EWT), etc.) at a resolution of 10 $\times$ 10 m. For instance, Fig. 17 exhibits a mapped Cab and LAI for Tunis parcel (Tun1). Fig. 18 displays an image of the same field taken from Google Earth in February 2020. Fig. 17(a) shows that Cab is at its maximum since in April the crop is at its full growth stage. But also, there are some green pixels in the border of the map and others in the middle, these pixels represent trees that border the field, as shown in Fig. 18. Unlike Cab, LAI is not homogeneous, and it indicates that the yield will be different from one subarea to another. This is due to several factors such as soil quality, the way of irrigation, etc. For instance, in [37], it has been demonstrated that the growth parameters and cereal yield are affected by the culture that preceded the sowing of cereals

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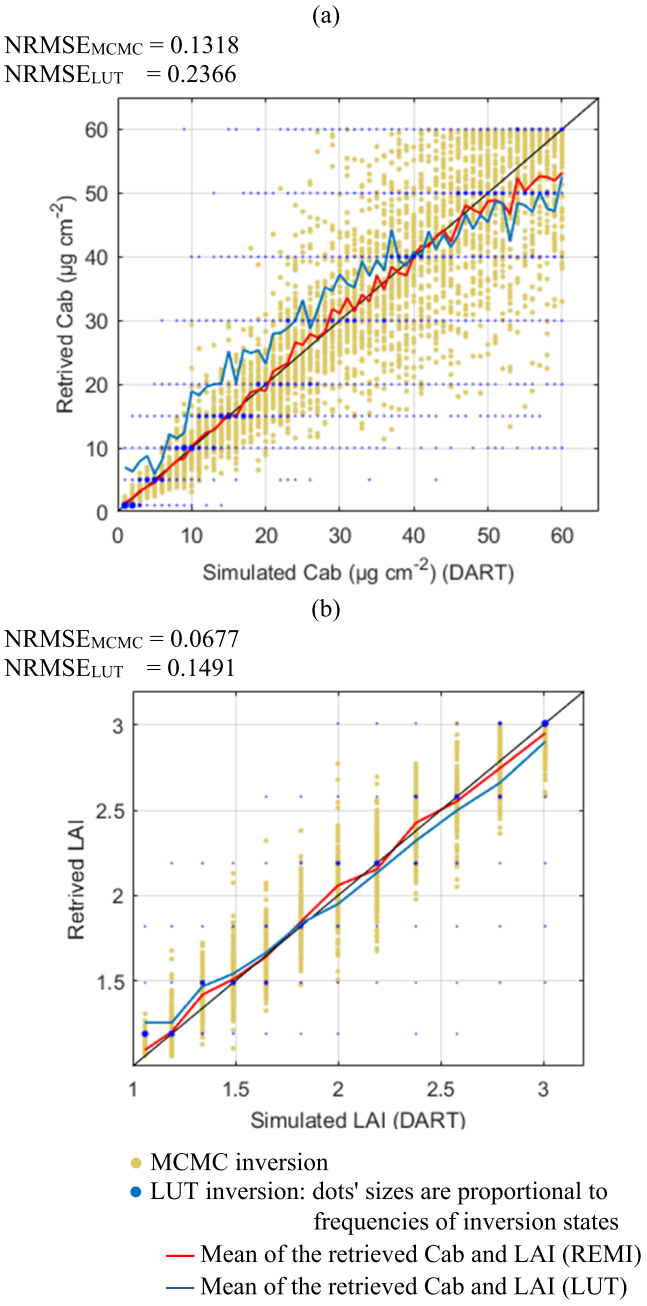

Fig. 16. REMI inversion performances versus noisy observation with a Gaussian signal $\epsilon \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 0.01$. (a) Retrieved Cab versus actual Cab. (b) Retrieved LAI versus actual LAI.

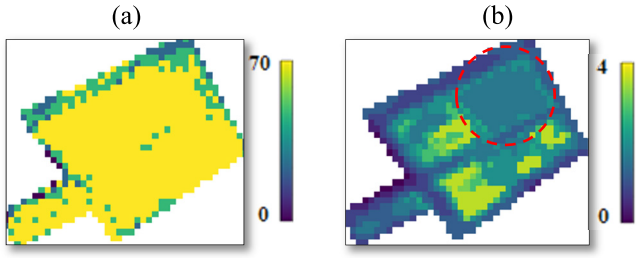

Fig. 17. Cab and LAI maps relative to Sentinel-2 image of TUNIS Parcel (Tun1) on 03 April 2021. (a) Cab (μg cm^{-2}) map. (b) LAI (m^2 m^{-2}) map.

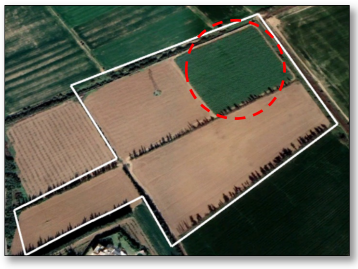

Fig. 18. TUNIS Parcel (Tun1) from Google Earth in February 2020.

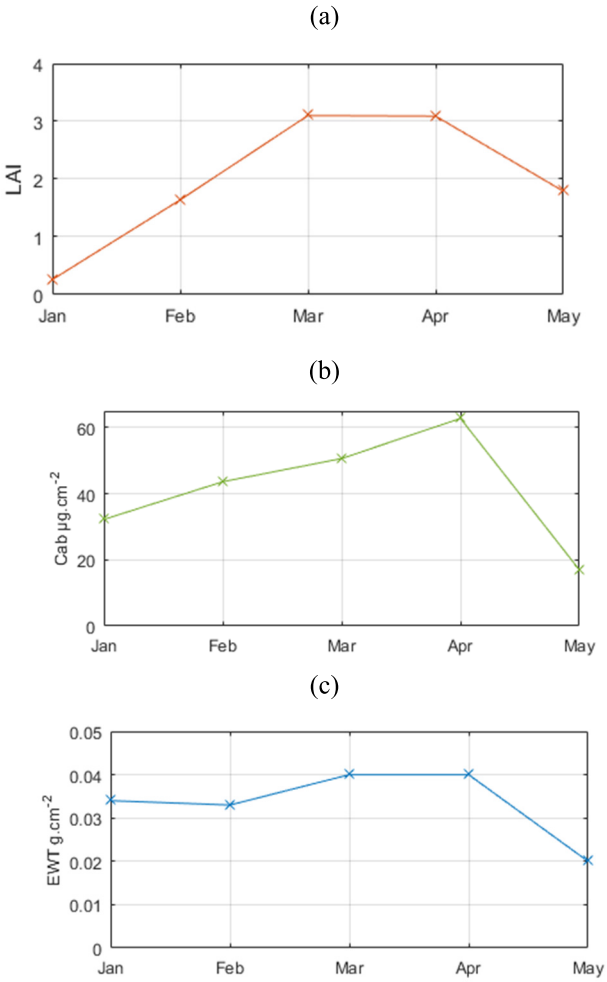

Fig. 19. Time variation of the retrieved crop properties of TUNIS Parcel (Tun1) from January to May 2021: (a) mean LAI, (b) mean Cab, and (c) EWT.

[37]. This process is called "rotation crop." Fig. 18 shows that this cultivation practice is applied to a restricted area (red circle) of the parcel (year 2020). Fig. 17(b) corresponding to the following year (2021) demonstrates that this process has induced a homogeneous and distinctive LAI of wheat cultivation in this area (red circle). Meanwhile, in the remain part of the parcel, LAI was heterogeneous. This confirms that the retrieved parameters, specifically here the LAI, is in accordance with the detailed practice in [37] and it remains dependent on the way the field is managed.

Moreover, general monitoring of the parcel throughout the crop's growth cycle is possible since Sentinel-2 images are available almost every week. Fig. 19 displays the monthly monitoring of the Tunis parcel (Tun1) for the year 2021. In January, the nonnull LAI value retrievals the new crop

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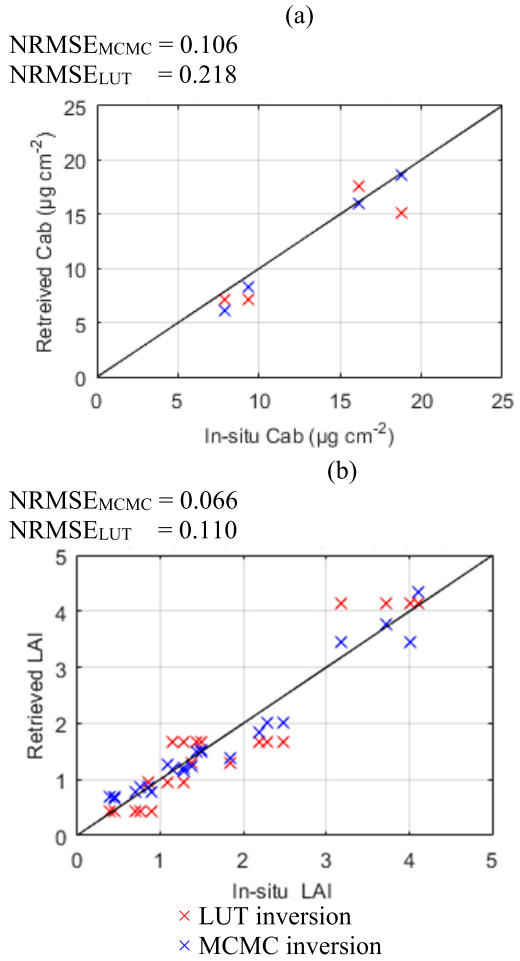

Fig. 20. Validation: (a) retrieved Cab versus in situ Cab. (b) Retrieved LAI versus in situ LAI.

growth. It will reach its maximum in March then stabilizes between March and April. We also observe that Cab and EWT contents practically follow this evolution. A small decrease in EWT is noted in February. The latter is due to the lack of rainfall between January and February also knowing that the farmer has not irrigated his crop in the meantime. In May, a strong loss of water content in the leaves is also recorded indicating that the crop is close to the harvesting period. At that time, leaves become dehydrated and turn to golden color. The latter also testifies a loss in the Cab content, as shown in the Cab curve; furthermore, LAI also shows a decrease in May. Indeed, when leaves lose water, they naturally roll up and shrink to reduce the surface exposed to the sun. Accordingly, this phenomenon will be observed by the satellite as being a reduction in the LAI. Our biophysical property retrieval is therefore consistent, which constitutes a good indication of the validity of our approach.

2) Validation: Fig. 20 illustrates the closeness between the retrieved LAI and Cab proprieties versus in situ measurements for the proposed technique and LUT. For both inversion techniques, we can see that the retrieved values are in agreement with in situ measurements; nevertheless, it can be shown that our approach provides the closest results to the 1:1 line. For instance, by observing Fig. 20(b) one can notice that all LAI in situ measurements upper than 3 are presented by a unique LAI$_{LUT}$ value equal to 4.2, while our proposed method seems to adapt well to these variations of this biophysical property. The results also reveal the accuracy of our method compared to LUT inversion in terms of NRMSE. For both LAI and Cab recovery, our method achieves a much better NRMSE than LUT method. Furthermore, Figs. 20(b) and 16(b) show that the LAI NRMSE computed for in situ measurements (NRMSE$_{MCMC}$ = 0.066) is close to the one computed for simulated data (NRMSE$_{MCMC}$ = 0.067). Also, from Fig. 20(a) and 16(a), we find close NRMSE values, 0.106 and 0.13, respectively. This means that the uncertainty in both cases is the same. In the simulated case, it is only equal to the observation noise standard-deviation, which is 0.01. The uncertainty in the actual case should be close to 0.01; however, it is not only due to the observation noise but also linked to the DART RTM imperfection. Indeed, in the simulated case, the RTM is assumed perfect since the used signatures in the inversion are those simulated by it, whereas in the actual case, the signatures are provided by the sensor and the RTM can just approximate such measurements since it neglects many effects.

V. Conclusion

In this study, we were interested in monitoring the growth cycle of field crops, particularly wheat and barley, using Sentinel-2 image time series. To do it, we inversed the 3-D RTM DART in the continuous domain (i.e., without variable sampling as in LUT) using a Bayesian technique. To overcome the DART long time running, we substitute an emulator for it. DART was employed to learn the emulator by simulating the Sentinel-2 images. For this task, two realistic mockups composed of soil, and 3-D wheat plants were created. The first mockup models the crop from started to grow to full-grown stage where the crops are particularly green. The second mockup has been designed with a 3-D wheat plant with curled leaf shapes that mark the end of the crop growth cycle where it ripens from green to a golden color. Besides, soil reflectances were extracted from Sentinel-2 images in the period after harvest and before sowing, that ensures the detection of pure soil pixels.

DART allows to generate a database of realistic reflectance of our mock-ups, that in turns we exploited it to learn and validate our emulator. The latter is based on residual artificial neural network (ResNet) architecture. This proposed architecture allowed us to achieve performances like RTM, yet, much faster. Besides, we showed that our emulator can react even to small variations in biophysical parameters (i.e., Cab μg cm^{-2} variation from 3 to 9 or even less).

In the inversion stage, the MCMC Bayesian technique was adapted with the goal of retrieving the most important parameters like Cab and LAI. It has the capacity to simulate statistical knowledge by an iterative processing of sampling such as the Metropolis sampler. We coupled our emulator to the MCMC algorithm and let the process of sampling converge to the a posteriori distribution of the biophysical properties and therefore retrieve their likely values.

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We validate our proposed approach with both simulated and in situ measurements of Cab and LAI. In the former case, the robustness against noisy observation was demonstrated. In the latter case, high accuracy for LAI and Cab retrieval was shown with NRMSE around 0.066 and 0.106, respectively. Therewith, our inversion method remains better than LUT inversion in all the cases.

The designed approach allowed field crops to be tracked from the start of growth to harvest from Sentinel-2 images. Monitoring of the biophysical parameters mapped at 10 m resolution can be performed with a revisit time of five days that corresponds to the Sentinel-2 satellites revisit period. Such high temporal and spatial resolution mapping can be coupled with crop model that allows best practice in water and nutrition management; moreover, our model can be adapted to monitor other crops such as olive and almond trees.

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